Generic Model to Estimate Dust Releases from Transfer/Unloading Operations of Solid Powders

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Introduction:

Under Section 5 of the Toxic Substances Control Act (TSCA), the U.S. Environmental Protection Agency's (EPA's) Office of Pollution Prevention and Toxics (OPPT) evaluates new chemicals (i.e., those chemicals not listed on the TSCA Inventory), for potential risks associated with their stated and potential uses. Existing chemicals may also be evaluated under Sections 4 and 6 of TSCA for potential risks associated with their various uses. In these cases, EPA may develop regulatory controls and/or non-regulatory actions to protect human health and the environment from harm resulting from manufacturing, processing, transport, disposal, and current and potential new uses of existing and new chemical substances.

A new chemical, with certain exceptions, is any chemical that is not currently on the TSCA Chemical Substance Inventory. Under Section 5 of TSCA, companies are required to submit a Premanufacture Notification (PMN) at least 90 days prior to commercial production (including importation) of a new chemical. The Chemical Engineering Branch (CEB) is responsible for preparing the occupational exposure and release assessments of the new chemicals. These assessments are based on information provided by the PMN submitter, information from readily available databases and literature sources, and standard estimating techniques used by CEB. Frequently, data on the new chemical being assessed are not available. In the event that information is unavailable, CEB relies on other approaches for developing release and exposure assessments.

CEB has developed a number of standard models to provide estimates of environmental releases and occupational exposures from standard release sources (e.g., equipment cleaning) and worker activities (e.g., unloading). These models are designed to provide conservative screening-level estimates where industry-specific or chemical-specific information is not available.

Scope:

This model estimates a loss fraction of dust that may be generated during the transferring/unloading of solid powders. While there are multiple potential industrial sources of dust (e.g., grinding, crushing), the scope of this model is limited to the transferring/unloading of solids. Specifically, this can be defined as activities where packaging/transport materials are opened and contents are emptied either into a feed system and conveyed or directly added into a process tank (e.g., reactor, mixing tank). The generic model is only applicable to estimate releases of solid powders lost during these activities. It does not cover solids lost from open reaction vessels or other downstream solids processing (e.g., drying, crushing, grinding). While these are potential sources of dust generation, a generic model for these sources may not be appropriate due to variations between industries and operations. This model does not estimate potential occupational exposures to the dusts generated during the aforementioned activities.

Approach:

In support of EPA, Eastern Research Group (ERG) completed Phase 1, Part 2 and Phase 2, Part 2, of the Generic Scenario Literature Search SOP (last revised June 13, 2006) as documented in Appendix A. Relevant articles/literature provided data to estimate a loss fraction of solids during transferring/unloading operations for a variety of industries. Please note that sources estimating dust emissions from mining operations and articles presenting theoretical approaches for estimating dust emissions were specifically excluded. Sources from various industries estimated the loss fraction during these operations to range from negligible to as high as 3 percent of the total unloaded/transferred material. Table 1 presents data used to determine the loss fraction. As available, information on the industry, activity, particle size, facility conditions, basis of estimate, media of release, and control technologies are included.

Some data were not used in the analysis because they contained emissions from additional activities outside the scope of the model. Table 2 summarizes these data. These data points were excluded for the following reasons:

- Estimate #1 in Table 2 included grinding operations, which are out of the scope of the model.
- Estimates #2 and #4 in Table 2 are dust release estimates for the total facility, not just for unloading/transfer activities.
- Estimate #3 in Table 2 included all raw mill (total facility) activities, which are outside the scope of the model.
- Estimate #5 in Table 2 included spray drying losses, which are outside the scope of the model.

Table 1. Summary Data Used to Estimate the Loss Fraction of Solids from Various Sources

Estimate Number	Industry	Activity	Estimated Loss Fraction	Notes	Media of Release / Control Technologies	Source/Basis
1	Liquid Coating Formulation	Unloading/handling solid raw materials	0.5%	Applicable to solid raw materials in solvent, water, wooden furniture, decorative paints, and auto coatings. Source references AP-42, 1983.	Releases initially to air then falls to shop floor. Cleaned up and sent to water, incineration or landfill.	OECD, 2006.
2	Automotive Coating Formulation	Unloading/pigment handling	2-3%	Based on DuPont Front Royal facility estimate of losses of pigment. Facility manufactured automobile refinishing coatings.	Collected in a filter and sent to off-site incineration.	Site visit report for the Latex/Emulsion Coatings Generic Scenario, 2006.
3	Latex/Emulsion Coating Formulation	Pigment handling	<0.1%	Based on McCormick Paint Fredrick facility estimate of losses of pigment. Facility manufactured water-based architectural coatings.	Collected by a filter and recycled into subsequent batches.	Site visit report for the Latex/Emulsion Coatings Generic Scenario, 2006.
4	Paint Formulation	Unloading/handling solid raw materials	0.54%	Based facility estimates of dust collection from a site visit to an OEM automotive paint formulator	Collected by a filter and sent to off-site incineration.	Environment Canada, 2003
5	Paint and Varnish Formulation	Pigment handling	0.5-1%	Based on engineering estimates from plant visits.	No media of release estimated.	AP-42, 1983.
6	Plastic Additives	Unloading/handling	0.2-0.6%	A worst case scenario for dust generation would be 0.6% for particle sizes less than 40 µm and 0.2% for particle sizes greater than 40 µm. Applicable to antioxidants, colorants, and stabilizers. Basis of estimate is unclear.	Release initially to air, then water or landfill due to particles settling.	OECD, 2004.

Table 1. Summary Data Used to Estimate the Loss Fraction of Solids from Various Sources (Continued)

Estimate Number	Industry	Activity	Estimated Loss Fraction	Notes	Media of Release / Control Technologies	Source/Basis
7	Plastic Additives (fillers)	Unloading/handling	0.1-0.5%	A worst case scenario for dust generation would be 0.5% for particle sizes less than 40 µm and 0.1% for particle sizes greater than 40 µm. Basis of estimate is unclear.	Release initially to air, then water or landfill due to particles settling.	OECD, 2004.
8	Printing Ink	Pigment handling	0.1%	Based on pigment handling/mixing. Based on engineering estimates from plant visits.	No media of release estimated.	AP-42, 1991.
9	Rubber Manufacturing	Unloading bags of raw materials	0.4%	Based on loss amount provided in a PMN submission (10 kg/s-d per 2,500 kg/s-d throughput). A fabric filter is used as the control technology (98.5% efficiency).	Used filters sent to incineration; uncaptured particulate releases assumed to air.	PMN submission.
10	Rubber Manufacturing	Loading (packaging) powdered material into bags	0.2%	Based on loss amount provided in a PMN submission (5 kg/s-d per 2,500 kg/s-d throughput). A fabric filter is used as the control technology (98.5% efficiency).	Used filters sent to incineration; uncaptured particulate releases assumed to air.	PMN submission.
11	Rubber Manufacturing	Loading (filling) powdered material into bulk bags	0.49%	Based on loss amount provided in a PMN submission (6 kg/s-d per 1,229 kg/s-d throughput). A fabric filter is used as the control technology (efficiency not provided).	Used filters sent to landfill.	PMN submission.
12	Misc. Chemical Manufacturing	Emptying raw material sacks	0.1%	Based on loss amount provided in a PMN submission (<1/1000 of PV).	Particulates from unloading area are washed into a sump and sent to WWT (assumed water release).	PMN submission.
13	Misc. Chemical Manufacturing	Unloading powdered raw materials	<0.1	Based on loss amount provided in a PMN submission (0.081%)	Particulates are collected via local exhaust and vented to atmosphere (assumed air release)	PMN submission.

Table 2. Summary Data Excluded from the Analysis

Estimate Number	Industry	Activity	Estimated Loss Fraction	Notes	Media of Release / Control Technologies	Source/Basis
1	Powder Coating Formulation	Unloading/handling solid raw materials and grinding	1.5%	Applicable to solid raw materials in powder coatings. Also includes dust generated from grinding activities. Source references 1994 Generic Scenario for Melt-Blend Processing of Powdered Coatings.	Releases initially to air then falls to shop floor. Cleaned up and sent to water, incineration or landfill.	OECD, 2006.
2	Concrete Manufacturing	All facility activities	0.006%	Estimate of uncontrolled facility emissions (captured emissions are excluded) but does not include road and wind blown dust.	No media of release estimated.	EPA FIRE Database, 2001.
3	Portland Cement Manufacturing	Raw material handling	0.0006%	Based on all raw mill operations.	Fabric filter used. No media of release estimated.	AP-42, 1995.
4	Inorganic Pigments Manufacturing	All facility activities	0.05%	Estimate of uncontrolled facility emissions (captured emissions are excluded).	No media of release estimated.	EPA FIRE Database, 1995.
5	Soap and Detergent Formulation	Spray drying, conveying, and loading	4.5%	Source notes that 85-99.9% of the dust is collected using APCD devices and recycled. Therefore, the actual release is 0.0045-0.675%.	No media of release estimated.	AP-42, 1991.

Model:

All data were reviewed to determine whether they were appropriate for this analysis. First, estimates not specific to the scope of the model were removed (i.e. estimates that included other processes), as previously discussed. After removing these data points, a cursory statistical evaluation was performed on data in Table 1.

Upper bound estimates (e.g., for Estimate #6, 0.6 percent was utilized) were taken for each industry and averaged. The average of the upper bound estimate for all industries is 0.4 percent. The median of the upper bound estimates is approximately 0.5 percent. Therefore, a conservative loss estimate would be 0.5 percent of the quantity transferred.

Additionally, laboratory-scale dust generation test data were reviewed. Although a theoretical model approach is outside the scope of this model, a study by Plinke, et al. investigated key parameters for developing a theoretical approach for estimating dust losses based on moisture content, particle size, drop height, and material flow (Plinke, 1995). Dust generation rates during unloading and transfers were measured for four materials. The highest measured dust generation rate was 0.5 percent. Although excluded from the above analysis, it provides further justification of a 0.5 percent loss fraction as a conservative estimate for the quantity of dust that may be released from transfer/unloading operations.

Based on these data, the following equation may be utilized to calculate a conservative, screening-level estimate of the quantity of dust that may be released from the unloading or transferring of solid powders:

$$Elocal_{dust\_generation} = Q_{chem\_transferred} \times F_{dust\_generation}$$

Where:

$Elocal_{dust\_generation}$	=	Daily release of dust from transfers/unloading (kg/site-day)
$Q_{chem\_transferred}$	=	Quantity of material transferred (kg/site-day)
$F_{dust\_generation}$	=	Fraction of chemical lost during transfers/unloading of solid powders (Default = 0.005 kg of released/kg handled)

Media of Release and Control Technologies:

Most facilities utilize some type of control device(s) to collect fugitive dust emissions. Many facilities collect fugitive dust emissions from these operations in filters and dispose of the filters in landfills or by incineration. Wet scrubbers may also be utilized by industry. However, in some cases, uncontrolled/uncollected particulates may be small enough to travel several miles from the facility, resulting in environmental and human exposures to the chemical of interest beyond the boundaries of the site. Fugitive dust emissions may also settle to facility floors and be disposed of when floors are cleaned (water if the floors rinsed or land or incineration if the floors are swept).

Therefore, the lost quantity of dust should conservatively be assessed as released to air, water, incineration, or landfill.

If the facility-specific information states a control technology is employed, the release may be partitioned to the appropriate media. If efficiency information is not available for the control technology, default control technology efficiencies are available in the CEB Engineering Manual (CEB, 1991). Table 3 provides estimated efficiencies for more common control technologies.

Table 3. Estimated Control Technology Efficiencies

Control Technology	Estimated Efficiency (%)	Notes/Source	Default Media of Release for Controlled Release
Filter (such as a baghouse)	> 99	For particles > 1 µm. CEB, 1991.	Incineration or Land
Cyclone/Mechanical Collectors:	80-99	For particles >15µm CEB, 1991.	Incineration or Land
Scrubber	Varies	Consult Table 7-1 of the CEB Engineering Manual.	Water

Therefore, the following equation may be used to estimate the portion of the release captured by the control technology:

$$\text{Elocal}_{\text{dust\_captured}} = \text{Elocal}_{\text{dust\_generation}} \times F_{\text{dust\_control}}$$

(to water, incineration, or landfill)

Where:

$\text{Elocal}_{\text{dust\_captured}}$ = Daily amount captured by control technology from transfers/unloading (kg/site-day)

$\text{Elocal}_{\text{dust\_generation}}$ = Daily release of dust from transfers/unloading (kg/site-day)

$F_{\text{dust\_control}}$ = Control technology capture efficiency (Default = 0 kg captured/kg processed; no control technlogy)

The following equation may be used to estimate the portion of the release that will not be captured by the control technology and may be released to air or settle the facility floor:

$$\text{Elocal}_{\text{dust\_fugitive}} = \text{Elocal}_{\text{dust\_generation}} \times \left(1 - F_{\text{dust\_control}}\right)$$

(to air, water, or landfill)

Where:

$\text{Elocal}_{\text{dust\_fugitive}}$ = Daily amount not captured by control technology from transfers/unloading (kg/site-day)

$\text{Elocal}_{\text{dust\_generation}}$ = Daily release of dust from transfers/unloading (kg/site-day)

$F_{dust\_control}$	=	Control technology capture efficiency (Default = 0 kg captured/kg processed; no control technlogy)

Uncertainties and Limitations:

Please note the following uncertainties and limitation of this model approach.

- Each estimate in Table 1 has a certain level of uncertainty, as discussed in Table 1.
- Each estimate presented in Table 1 is from a different data source, and each estimate may have been generated in a slightly different manner. This approach assumes that each estimate presented in Table 1 represents the total dust generation that may or may not be captured by a control technology.
- As additional PMN submission data or additional industry data become available, this loss estimate may be updated as appropriate.
- This approach is designed for screening-level estimates where appropriate industry-specific or chemical-specific information is not available.

Sample Calculations:

Unknown Control Technology

A PMN submission states 1,000 kg of solid powder are unloaded at a site each day. The total amount of dust released during unloading would be:

$$Elocal_{dust\_generation} = Q_{chem\_transferred} \times F_{dust\_generation}$$

$$Elocal_{dust\_generation} = 1{,}000 \frac{kg}{site\text{-}day} \times 0.005$$

$$Elocal_{dust\_generation} = 5 \frac{kg}{site\text{-}day}$$

(to air, water, incineration or landfill)

Known Control Technology

A PMN submission also states that a baghouse filter (99 percent efficiency) is used to collect fugitive dust particles during unloading activities. This filter is sent to incineration or landfill. The total amount of substance captured by the control technology would be:

$$Elocal_{dust\_captured} = Elocal_{dust\_generation} \times F_{dust\_control}$$

$$Elocal_{dust\_captured} = 5 \frac{kg}{site\text{-}day} \times 0.99$$

$$\text{Elocal}_{\text{dust\_captured}} = 4.95 \frac{\text{kg}}{\text{site - day}}$$

(to incineration or landfill)

Then, the fugitive release of the substance (not captured by the baghouse filter) would be:

$$\text{Elocal}_{\text{dust\_fugitive}} = \text{Elocal}_{\text{dust\_generation}} \times \left(1 - \text{F}_{\text{dust\_control}}\right)$$

$$\text{Elocal}_{\text{dust\_fugitive}} = 5 \frac{\text{kg}}{\text{site - day}} \times (1 - 0.99)$$

$$\text{Elocal}_{\text{dust\_fugitive}} = 0.05 \frac{\text{kg}}{\text{site - day}}$$

(to air, water, or landfill)

References:

(AP-42, 1991) U.S. EPA AP-42, Volume I, Fifth Edition, January 1995. http://www.epa.gov/ttn/chief/ap42/#drafts.

(AP-42, 1983) U.S. EPA AP-42, Volume I, Fifth Edition, January 1995. http://www.epa.gov/ttn/chief/ap42/#drafts.

(CEB, 1991) *CEB Manual for the Preparation of Engineering Assessment, Volume 1*. U.S. EPA; Office of Pollution Prevention and Toxics; Chemical Engineering Branch; Washington DC; Contract No. 68-D8-0112; February 1991.

(Environment Canada, 2003) Environment Canada. Release Scenarios for Canadian Coatings Facilities, January 24, 2003 draft.

(EPA FIRE Database, 2005) U.S. EPA Factor Information REtrieval (FIRE) Database. Version 6.25, October, 18, 2004. http://www.epa.gov/ttn/chief/software/fire/index.html.

(EPA FIRE Database, 2001) U.S. EPA Factor Information REtrieval (FIRE) Database. Version 6.25, October, 18, 2004. http://www.epa.gov/ttn/chief/software/fire/index.html.

(OECD, 2006) Organisation for Economic Co-operation and Development (OECD). OECD Series on Emission Scenario Documents (ESDs). Draft ESD on Coatings Industry (Paints, Lacquers, and Varnishes), June 2006.

(OECD, 2004) Organisation for Economic Co-operation and Development (OECD). OECD Series on Emission Scenario Documents (ESDs), Number 3. ESD on Plastic Additives, June 2004. http://www.olis.oecd.org/olis/2004doc.nsf/LinkTo/env-jm-mono(2004)8.

(Plinke, 1995) Plinke, Marc A.E., et al. Dust Generation from Handling Powders in Industry. *American Industrial Hygiene Association Journal*. Vol. 56: 251-257, March 1995.

(SVR for Latex/Emulsion Coatings Generic Scenario, 2006) U.S.EPA, CEB. Site Visit Report for DuPont Performance Coatings - Front Royal, Virginia facility. Information gathered for the Draft Generic Scenario: Formulation of Latex/Emulsion Coatings. 2003.

(SVR for Latex/Emulsion Coatings Generic Scenario, 2006) U.S.EPA, CEB. Site Visit Report for McCormick Paint Works Company – Frederick, Maryland facility. Information gathered for the Draft Generic Scenario: Formulation of Latex/Emulsion Coatings. 2003.

Appendix A: Literature Search Documentation Table

Generic Scenario Literature Search Documentation Table
(Generic Scenarios Development Project SOP, June 13, 2006 version)

Phase 1: Standard Literature Search
Researchers: Tracey Pennington/Daryl Hudson
Phase 1 Completion Date: September 20, 2006

Primary Keywords: Dust, Dust Releases, Dust Emissions, Particulate, Transfer Operations, Unloading

Phase 1, Part 2: Search of Standard CEB Sources

Source	Search Description	Results
U.S. EPA CEB	Followed GS Literature Search SOP	Loss fraction estimate from supporting documentation for the draft Generic Scenario for Latex Coatings.
U.S. EPA TRI	N/A	
U.S. EPA Office of Water	N/A	
U.S. EPA Office of Air	Followed GS Literature Search SOP	2 articles describing National Emission Standards for Lime Manufacturing and Cement Manufacturing plants. Information characterized overall emissions, however, no information on raw material unloading was available. 2 articles describing estimation of air emissions for Paint, Ink and Chemical Manufacturing Facilities. Information characterized emission estimates of VOSs, not particle emissions.
U.S. EPA OECA Sector Notebooks	Followed GS Literature Search SOP	Notebook for Profile of Stone, Clay, Glass and Concrete industry Information characterized overall emissions from each industry. No information was provided for raw material unloading.
U.S. EPA AP-42	The following industries were searched: synthetic rubber, synthetic fibers, soap and detergents, printing inks, paint and varnishes, carbon black, lime manufacturing, Portland cement, and ceramic products manufacturing.	Emission factors were found for paints and varnishes, printing ink, and Portland cement.

Phase 1, Part 2: Search of Standard CEB Sources (Continued)

Source	Search Description	Results
Other U.S. EPA (e.g., DfE)	Followed GS Literature Search SOP	No relevant documents found.
SRI	N/A	
OSHA	Followed GS Literature Search SOP	2 articles regarding workplace exposure of particulates; however, articles did not contain information on environmental releases.
NIOSH	Followed GS Literature Search SOP	No relevant documents found.
OECD	The following industries were searched: plastic additives, coating formulations, painting via spraying, lubricants and lubricant additives. Emission factors were found for paints and varnishes, printing ink, and Portland cement.	Several estimates for particulate losses were found in plastic additives and coating formulations.
Environment Canada	Followed GS Literature Search SOP	No relevant documents found.
Canadian P2 Information Clearinghouse	Followed GS Literature Search SOP	No relevant documents found.
U.S. Census Bureau	N/A	
NC Division of Pollution Prevention and Environmental Assistance	Followed GS Literature Search SOP	No relevant documents found.
Kirk-Othmer	Followed GS Literature Search SOP	No relevant documents found.
Other Sources (EPA FIRE Database)	Followed GS Literature Search SOP	Two estimates for total solid/particulate releases from cement manufacturing and inorganic pigment manufacturing facilities were found. However, there was no breakout of individual sources/activities.

Phase 2: Targeted Literature Search
Researcher: Tracey Pennington
Phase 2 Completion Date: September 21, 2006

Keywords and Search Strategy:
Search Strategy: 1) Identified search databases. (2) Entered Primary Keyword Search: Dust Generation (3) Utilized Secondary Keywords Search to try to identify additional sources missed from primary search: Dust Emissions, Unloading, Dust Release; (4) Google search of specific industries that might have relevant information
Internet Search Tool: www.google.com

Phase 2, Part 2: Abstract Database Search

Abstract Database	Search Description	Location of Results
National Technical Information Service	Followed Search Strategy above.	Q:\CEB\Generic Scenarios\Dust Model\Dust Model Abstract.pdf. Abstract for Fugitive Dust Model User's Guide. Description of specific content of document is vague. Uncertain if document is specific to raw material unloading emissions.
SCIRUS	Followed Search Strategy above.	Q:\CEB\Generic Scenarios\Dust Model\SCIRUS Dust #1 through Q:\CEB\Generic Scenarios\Dust Model\SCIRUS Dust #4.pdf
www.google.com	Followed Search Strategy above.	Q:\CEB\Generic Scenarios\Dust Model\Falling Powder Equation.pdf This document provides a theoretical model for estimation of specific generation rate of respirable dust. This was not within the scope of the model.